Robotic Systems in Manipulation 2025/2026

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Laboratory Project: KUKA LBR MED

Mestrado em Engenharia Informática e de Computadores

The proposed work consists in the Kinematic and Dynamic analysis of the *KUKA LBR MED* surgical manipulator. In your report, provide complete, concise answers to the points below, using your own code. For developing your computational models, a Symbolic Robotics Matlab Toolbox is provided in the course webpage, as the starting point.

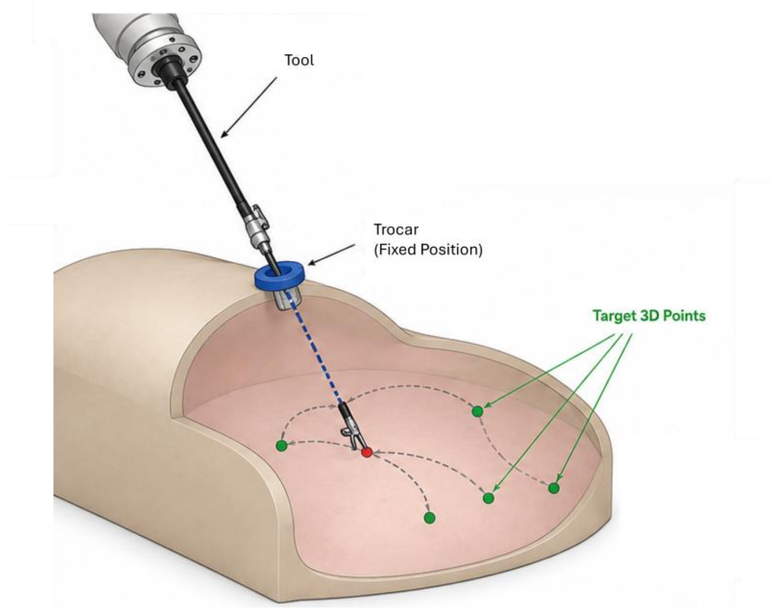
Kinematics: (Due on May 17th)

1. Analyze the robot's drawing and build its kinematic model according to the Denavit-Hartenberg convention. To do so, draw your own joints diagram with the link reference frames and the corresponding table of parameters.
2. Using the provided Toolbox, create a Simulink model for the direct kinematics of the robot. Validate your model by placing the robot in configurations for which the end-effector's position and orientation can be easily determined.
3. Following a kinematic decoupling approach present one closed-form solution for the inverse kinematics of the robot and identify the singularity conditions. Create the Simulink model and validate your model against the direct kinematics model implemented in Point 2. (NOTE: the manipulator is redundant)
4. Create a Simulink model for the geometric Jacobian of the robot. Validate the result through numerical differentiation of the direct kinematics in Point 2. Show the effect of the kinematic singularities of the robot on the rank of the Jacobian matrix.
5. Using the direct kinematics and the geometric Jacobian models, determined in Points 2 and 4, implement a Simulink model for the Closed Loop Inverse Kinematics of the robot, stabilizing the null space. Validate your model against the results of Point 3.

Trajectory execution: (Due on June 7th)

Consider that the KUKA MED robot is attached with a surgical (rigid) tool mounted at the end-effector (e.g., a biopsy needle). The tool (length, L=15cm) is inserted in the body through a **trocar**, which has a **fixed position**, and the tooltip must reach the desired targets while not violating the constraint imposed by the trocar.

For a given list of targets m_i and a trocar position c_r, plan a trajectory that places the tooltip at each target. Validate on the simulated robot.



Note on Matlab toolboxes

To verify that you have the necessary Matlab and Simulink toolboxes, you may execute the following commands in Matlab's command window.

```
>> syms x
>> y=x^2
>> new_system('model_name')
>> open_system('model_name')
>> matlabFunctionBlock('model_name/parabola',y)
```